

Shruti A. Panicker

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Seeking full-time opportunities – Available Immediately

SUMMARY

AI/ML practitioner and Data Scientist with 4+ years of experience applying statistical modeling, machine learning, experimentation, and behavioral analytics to improve customer engagement and inform product strategy. Led data science and analytics initiatives for a Fortune 20 technology client, transforming large-scale customer and product-usage data into executive decisions, growth opportunities, KPI frameworks, and product-roadmap recommendations. M.S. in Artificial Intelligence with research experience in NLP, LLM evaluation, and Voice AI.

EDUCATION

M.S. in Artificial Intelligence – ML Concentration, Northeastern University *Boston, MA, USA* | Sep 2024 – May 2026
GPA: 3.875/4 | **Coursework:** Foundations of AI, Data Processing for AI, Machine Learning, NLP, Human Computer Interaction
B.Tech in Computer Science and Engineering, Amrita Vishwa Vidyapeetham *Coimbatore, India* | Jul 2016 – Aug 2020
GPA: 8.06/10 | **Coursework:** Data Structures, Computational Intelligence, Deep Learning, Data Science

EXPERIENCE

Graduate Teaching Assistant, Northeastern University *Boston, MA, USA* | Sep 2024 – Apr 2026

- Led instruction, office hours, assignment design, and project evaluation for DS2500 across four semesters, supporting 250+ students per semester.

Decision Scientist Lead, Mu Sigma Business Solutions *Bangalore, India* | Jan 2024 – May 2024

- Led a 10-person decision science team delivering SaaS product analytics for a Fortune 20 technology client's global collaboration-device portfolio, translating behavioral insights into executive roadmap decisions presented to C-suite leadership.
- Built a statistical upsell propensity model in Python and SQL, applying TAM/SAM/SOM segmentation across large-scale datasets to identify high-conversion targets and **drive 5% product adoption growth** through focused campaigns.
- Set fiscal year targets for 5+ OKRs using time series forecasting, establishing a statistical target-setting framework where none existed.
- Built and automated 25+ product KPI dashboards in Power BI using Python and SQL, eliminating manual data extraction and enabling weekly decision-making for product leadership.

Decision Scientist, Mu Sigma Business Solutions *Bangalore, India* | Sep 2020 – Dec 2023

- Built a graph-based entity resolution system to deduplicate Device IDs across large-scale datasets, applying network association analysis to identify anomalous identity patterns and restore integrity across all downstream portfolio metrics.
- Designed a three-state customer lifecycle model (inactive/churned/reactivated) using classification and behavioral segmentation, enabling targeted re-engagement campaigns that **drove 18% engagement increase within 6 months**.
- Diagnosed product onboarding drop-off by joining behavioral usage signals with support data in Python and SQL, pinpointing friction points across onboarding stages and recommending targeted interventions that **drove a 30% increase in reactivation rates**.
- Led an 8-month EUDB-compliant pipeline migration across 25+ dashboards, redesigning ETL transformation and aggregation layers to ensure EU data residency compliance with zero loss in metric accuracy.

Data Analyst, Riveron Infotech Pvt Ltd *Trivandrum, India* | Jan 2020 – Aug 2020

- Conducted cohort and customer-journey analyses to identify progression and engagement drop-offs, contributing to targeted interventions that **improved program completion by 18%**.
- Built and analyzed a 6-stage website-to-paid-enrollment funnel using Python, SQL, and Tableau across 5K+ sessions, identifying highest-friction drop-off points and recommending UX improvements to **lift paid enrollment conversion by 15%**.
- Developed a churn prediction model using behavioral signals (login frequency, activity completion, mentor engagement) to flag at-risk students, enabling proactive outreach that **improved retention by 30%**.

PROJECTS

Semantic Turn Detection for Voice AI *Northeastern University*
Built an intent-conditioned turn-detection system using a Groq-hosted Llama-3.3-70B ensemble, classifying utterances by intent to dynamically adjust response timing. Designed annotation guidelines and a within-subject evaluation with 18 participants; compared with a silence-only VAD baseline, improved naturalness from 4.31 to 6.00 and reduced frustration from 3.08 to 1.53 on seven-point Likert scales.

Empathy Detection – WASSA 2024 Shared Task *Northeastern University*
Fine-tuned BERT regression models on 11K+ conversational samples to predict empathy, emotional polarity, and emotional intensity. Designed intrinsic and extrinsic evaluation frameworks and achieved a 0.597 average Pearson correlation, exceeding the published score of the fourth-ranked WASSA 2024 submission.

SKILLS

Languages & Query: Python, SQL, R, C# | **ML & Modeling:** Pandas, NumPy, PyTorch, TensorFlow, Keras, scikit-learn, Statistical Modeling | **NLP & LLMs:** BERT, Transformers, Text Classification, Sentiment Analysis, LLM Evaluation | **Data & Analytics:** A/B Testing, User Behavior Analysis, Cohort & Funnel Analysis, Time Series Forecasting | **Data Engineering:** PySpark, AWS, Amazon SageMaker, Azure, Databricks, Hadoop | **Visualization & BI:** Power BI, Tableau | **Certifications:** AWS Certified Cloud Practitioner, AWS Certified Machine Learning Engineer – Associate